Coding & Solution

Date	04 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/akhilreddy772/CreditCardApprovalPrediction.git
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, XGBoost, Scikit-learn, Pandas, NumPy
Key Features Implemented	User Registration and Login Authentication • Credit Card Approval Prediction using XGBoost Model • Applicant Information Collection through Dynamic Forms • Automatic Monthly Income Calculation • Prediction Result with Confidence Score • Prediction History Management • Responsive Banking Dashboard UI • Secure Session Management
Pending / Incomplete Features	• Email Notification for Prediction Results • Database Integration for Permanent Storage (Currently Session-Based) • Admin Dashboard for Monitoring Users • Cloud Deployment (AWS/Render/Heroku) • Explainable AI (SHAP/LIME) for Detailed Prediction Explanation
Setup / Run Instructions	**Step 1:** Install Python 3.10 or above. **Step 2:** Clone or download the project repository. **Step 3:** Create a virtual environment. **Step 4:** Install all dependencies using pip install -r requirements.txt. **Step 5:** Ensure the trained XGBoost model (xgboost_credit_model.pkl) is available in the project folder. **Step 6:** Run the application using python app.py. **Step 7:** Open the browser and visit http://127.0.0.1:5000. **Step 8:** Register a new account or log in using existing credentials. **Step 9:** Enter applicant details and click **Evaluate Application Risk** to view the prediction result.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Credit Card Approval Prediction System follows a modular architecture where the frontend, backend, and machine learning model are integrated efficiently. The Flask framework manages user authentication, request routing, session handling, and communication between the user interface and the trained XGBoost model. The machine learning model processes applicant information after preprocessing and generates an instant approval or rejection decision with a confidence score.

The application emphasizes clean coding practices by separating templates, static resources, backend logic, and model files into dedicated directories. Proper input validation, exception handling, reusable functions, and organized project structure improve maintainability and scalability. The project can be further enhanced by integrating a relational database, implementing cloud deployment, adding explainable AI techniques, and enabling administrator-level monitoring for real-world banking environments.